

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1511

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**
Question Count: 3

Code of Conduct

I **V.PREETE (192572004)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V.PREETE

Case Study 1:

Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

ANSWER:

Overview

A retail chain may have many stores located in different places. Managing products, stock levels, sales transactions, and customer demand manually can be difficult. A cloud-based inventory management system provides a centralized solution where all stores can access and update inventory information in real time.

The system connects Point-of-Sale (POS) systems, store devices, dashboards, cloud services, databases, and analytics platforms. Store employees can check stock availability, while managers can monitor sales and inventory through dashboards.

Role of Cloud Components

1. Clients

POS systems and dashboards act as clients. POS systems record sales and update inventory whenever a product is purchased. Managers use dashboards to view stock levels, sales reports, and product information.

2. APIs

APIs provide communication between POS systems and cloud services. For example, when a product is sold, the POS system can send the transaction to the cloud through an API.

The API can perform operations such as:

- Add product information

- Update stock quantity
- Record sales
- Retrieve inventory
- Generate reports

3. Cloud Storage

Cloud storage stores large amounts of information such as:

- Product details
- Transaction records
- Sales history
- Product images
- Analytics datasets
- Backup information

Cloud storage provides scalability because storage capacity can be increased when the amount of data grows.

4. Platform Services

Platform services are used for analytics and demand forecasting. Historical sales data can be analyzed to predict future demand.

For example, if a particular product sells more during a festival season, the system can forecast higher demand and recommend increasing stock.

5. Security

Security is important because the system handles business and customer information. Security can be implemented using:

- VPN
- Firewalls
- API gateways
- Authentication
- Authorization
- Encryption
- Role-based access control

Only authorized employees should be allowed to access inventory and business information.

Benefits

The system provides real-time inventory visibility across all stores. It reduces manual work and helps prevent overstocking and stock shortages. Cloud-based analytics can improve demand forecasting and help managers make better decisions.

It also provides scalability because additional stores and users can be added without completely redesigning the system.

2. Case Study 2:

Cloud-Based Document Collaboration System An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

ANSWER:

Overview

A cloud-based document collaboration system is similar to applications such as Google Docs. It allows multiple users to create, edit, share, and manage documents through web browsers.

Instead of storing documents only on a local computer, documents are stored in distributed cloud storage. Users can access their documents from different devices and locations.

Role of Cloud Components

1. Web Clients

Users access the application using browsers or other supported clients. They can create new documents, edit existing documents, and share documents with other users.

For example, two employees can work on the same document from different locations.

2. Web APIs

Web APIs allow different clients to communicate with cloud services. When a user makes a change to a document, the change is sent to the cloud through an API.

APIs are also responsible for:

- Document creation
- Document retrieval

- Sharing
- Synchronization
- User management
- Saving changes

3. Distributed Cloud Storage

Documents are stored in distributed cloud storage. Data can be replicated across multiple servers or locations.

This improves:

- Availability
- Reliability
- Data durability
- Disaster recovery

If one server becomes unavailable, another copy of the data can be accessed.

4. Version Control

Version control maintains different versions of documents. If a user accidentally deletes or changes important content, an earlier version can be restored.

For example:

Document Version 1

↓

Document Version 2

↓

Document Version 3

↓

Current Version

This is useful for businesses where documents are frequently modified by several employees.

5. Network Infrastructure

A strong network infrastructure is required to provide low latency and high availability. Content and services can be distributed across different geographical locations.

This allows users from different countries or regions to access documents quickly.

6. Security

Documents may contain confidential business information. Therefore, strong security mechanisms are required.

Security includes:

- Data encryption
- Identity management
- User authentication
- Access permissions
- Role-based access control
- Secure communication using HTTPS

For example, a document owner may give another employee **view-only access** instead of editing permission.

Benefits

The cloud-based collaboration system allows users to work from anywhere. Multiple users can collaborate on documents without exchanging files through email or USB devices.

It reduces dependency on local storage and provides automatic backup, version control, scalability, and availability.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

ANSWER:

Overview

An online food ordering platform allows customers to browse restaurants and menus, select food items, place orders, and track their orders using web browsers or mobile applications.

The frontend communicates with backend services using RESTful Web APIs. Cloud infrastructure such as virtual machines and managed Kubernetes provides scalability and reliability.

Role of Cloud Components

1. Frontend Clients

Customers use web browsers or mobile applications to access the platform.

The frontend provides features such as:

- Restaurant search
- Menu browsing
- Food selection
- Cart management
- Order placement
- Order tracking
- Customer account management

2. RESTful Web APIs

REST APIs connect the frontend with backend cloud services.

For example:

Mobile App



POST /orders



Backend Service



Order Database

APIs can handle login, restaurant information, menu details, orders, payments, and delivery information.

3. Cloud Storage

Cloud storage is used for storing objects such as:

- Menu images
- Restaurant images
- Invoices
- Customer documents
- Application files
- Backup data

Cloud storage can handle large amounts of data and can be expanded when required.

4. Virtual Machines

Virtual machines provide computing resources for running backend applications and services.

They allow the organization to create multiple computing instances based on demand.

5. Managed Kubernetes

Kubernetes manages containers and helps the application scale automatically.

For example, during lunch or dinner time, the number of users may increase significantly.

Normal Traffic



Few Containers

High Traffic



More Containers



Better Performance

Kubernetes can distribute workloads between containers and replace failed containers.

6. Security

Security is essential because the platform handles customer accounts, orders, and potentially payment-related information.

Security mechanisms include:

Authentication

Tokens are used to verify authenticated users.

HTTPS:

HTTPS encrypts communication between clients and servers.

Role-Based

Different users receive different permissions.

For example:

User	Access
Customer	Place and view orders
Restaurant	Manage menu and orders
Delivery Partner	View assigned deliveries

User	Access
Admin	Manage complete system

Benefits

The cloud-based food ordering platform can support a large number of users. Kubernetes and cloud infrastructure provide scalability and reliability. REST APIs make communication between different application components simple.

Cloud storage provides secure and scalable data storage, while authentication and access control protect user information.